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RA - Q1 (PIZZA)

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Q1. $\pi_{\text{name}} (\sigma_{\text{age} < 18} (\text{Person}))$

Q2. $\pi_{\text{DISTINCT pizzeria, pizza, price}} (\sigma_{\text{name} = \text{"Amy"}} (\text{Eats} \bowtie \text{Server}))$
 $\wedge \text{price} < 10$

Q3. $\pi_{\text{pizzeria, name, age}} (\sigma_{\text{age} < 18} (\text{Person} \bowtie \text{Frequent}))$

Q4. $\pi_{\text{pizzeria}} (\text{Frequent} \bowtie \sigma_{\text{age} < 18} (\text{Person}))$
 $\wedge$

$\pi_{\text{pizzeria}} (\text{Frequent} \bowtie \sigma_{\text{age} > 30} (\text{Person}))$

Q6. $\pi \left(\begin{array}{c} \text{name,} \\ \text{total Pizza} \end{array} \left(\begin{array}{c} \sigma_{\text{total Pizza} \geq 2} \left(\begin{array}{c} \gamma_{\text{name,}} \\ \text{COUNT(pizza)} \rightarrow \text{total Pizza} \end{array} \right) \end{array} \right) \right)$

total Pizza ↓

Q7. $\pi_{\text{avgPrice}} \left(\begin{array}{c} \text{pizza,} \\ \text{avgPrice} \end{array} \left(\begin{array}{c} \gamma_{\text{pizza,}} \\ \text{AVG(price)} \rightarrow \text{avgPrice} \end{array} \right) \right)$

avgPrice ↓

Q3.

Let $A = \text{Person}$, $B = \text{Frequent}$, $C = \text{Frequent}$,
 $D = \text{Person}$

$$\text{Answer} = \pi_{\substack{B.\text{pizzeria} \\ A.\text{name} \\ A.\text{age}, \\ D.\text{name}, \\ D.\text{age}}} \left(\sigma_{\substack{A.\text{age} < 18 \\ A \\ D.\text{age} > 30}} \left(A \bowtie B \bowtie C \bowtie D \right) \right)$$

$$\begin{array}{ccc} A & B & C & D \\ A.\text{name} & B.\text{pizzeria} & C.\text{name} & \\ = & = & = & \\ B.\text{name} & C.\text{pizzeria} & D.\text{name} & \end{array}$$